

Exchange-correlation functional with broad accuracy for metallic and nonmetallic compounds, kinetics, and noncovalent interactions

Yan Zhao, Nathan E. Schultz, and D. G. Truhlar

Citation: *J. Chem. Phys.* **123**, 161103 (2005); doi: 10.1063/1.2126975

View online: <http://dx.doi.org/10.1063/1.2126975>

View Table of Contents: <http://jcp.aip.org/resource/1/JCPSA6/v123/i16>

Published by the [American Institute of Physics](#).

Additional information on J. Chem. Phys.

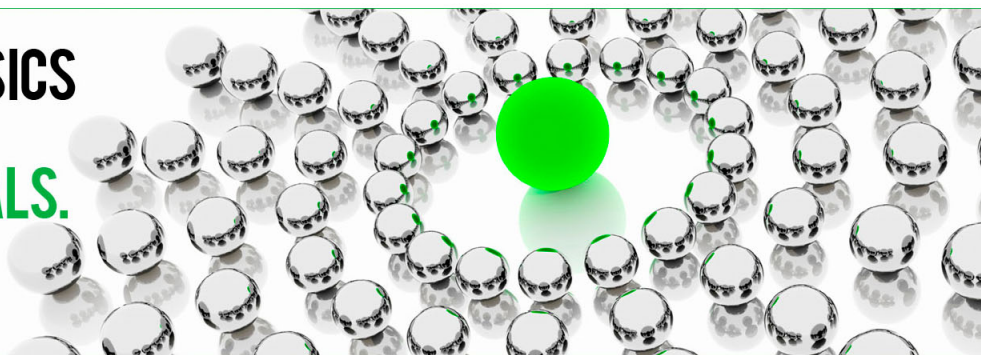
Journal Homepage: <http://jcp.aip.org/>

Journal Information: http://jcp.aip.org/about/about_the_journal

Top downloads: http://jcp.aip.org/features/most_downloaded

Information for Authors: <http://jcp.aip.org/authors>

ADVERTISEMENT



**ALL THE PHYSICS
OUTSIDE OF
YOUR JOURNALS.**

physics
today

www.physics today.org

Exchange-correlation functional with broad accuracy for metallic and nonmetallic compounds, kinetics, and noncovalent interactions

Yan Zhao, Nathan E. Schultz, and D. G. Truhlar^{a)}

Department of Chemistry and Supercomputing Institute, University of Minnesota, 207 Pleasant Street S.E., Minneapolis, Minnesota 55455-0431

(Received 19 August 2005; accepted 29 September 2005; published online 24 October 2005)

By incorporating kinetic-energy density in a balanced way in the exchange and correlational functionals and removing self-correlation effects, we have designed a density functional that is broadly applicable to organometallic, inorganometallic, and nonmetallic bonding, thermochemistry, thermochemical kinetics, and noncovalent interactions as well as satisfying the uniform electron gas limit. The average error is reduced by a factor of 1.3 compared with the best previously available functionals, but even more significantly, we find a functional that has a high accuracy for all four categories of interaction. © 2005 American Institute of Physics. [DOI: 10.1063/1.2126975]

Although the density-functional theory (DFT) of electronic structure was developed 40 years ago, progress was slow until the development of modern functionals in the last 20 years.^{1,2} Now density-functional theory is the method of choice for most applications involving metals, catalysis, dynamics, nanotechnology, and large or complex molecules. The key to progress is the design of new functionals incorporating correct mathematical properties and improved physical descriptions of exchange and correlation^{3–16} and stringently tested against well-validated, diverse, and representative data. Although a few density functionals are determined entirely from first principles, most successful functionals in chemistry have the values of one or more parameters determined by fitting.^{4,5,7–9,12–14,17–28} None of the available functionals, though, is universally accurate. The variables used to construct most functionals are the local spin density ρ_σ , its gradient $\nabla\rho_\sigma$, and two quantities that can be constructed from the Kohn-Sham orbitals: Hartree-Fock (HF) exchange and, most recently, kinetic-energy density, represented here by the variable called τ_σ which equals the kinetic-energy density for spin σ , where $\sigma=\alpha$ (up) or β (down).

Local-spin-density approximations lead to a significant overbinding of molecules and solids.¹ Although this is largely corrected by including $\nabla\rho_\sigma$ for metals or including $\nabla\rho_\sigma$ and 20%–25% HF exchange for molecules, chemical-reaction-barrier heights are still systematically underestimated. The results can be improved by using ~40% HF exchange.²² However, including even much smaller amounts of Hartree-Fock exchange (>10%) degrades the accuracy of many metal-metal²⁹ and metal-ligand³⁰ bond energies. Thus for systems such as metalloenzymes, heterogeneous and homogeneous catalysts involving metals or metal compounds, and nanotechnological devices with metalorganic components, one cannot achieve a uniformly satisfactory description of all bonds in the system. Furthermore, only very recently have functionals been obtained that give a satisfactory

practical accuracy for weak dispersionlike interactions.^{19,27,28}

Many different aspects of density functionals have been singled out in trying to understand these performance issues. In wave-function theory (WFT), both electron exchange and electron correlation require at least two electrons, but most density functionals include artificial and incorrect self-exchange and self-correlation. Since kinetic-energy density, in conjunction with $\nabla\rho_\sigma$, can “detect” one-electron regions,¹³ one hopes it can be used to ameliorate these problems. However, so far, kinetic-energy density has usually been included separately in either the exchange^{24,26} or the correlation^{9,28} functional, whereas a more balanced treatment may be required.¹⁵ This requirement is signaled by the fact that DFT exchange potentials include some effects that appear in WFT as correlation,^{14,31} in particular, static correlation (also called near-degeneracy correlation). It is sometimes stated that one could include a larger fraction of self-exchange-free Hartree-Fock exchange if one had a more accurate correlation functional,¹⁶ but with an inaccurate correlation functional, one cannot replace local exchange by Hartree-Fock exchange because most correlation functionals are more compatible (due to cancellation of errors) with local exchange calculated from ρ_σ and $\nabla\rho_\sigma$ than with HF exchange.^{2,16,31} In the present work we present a more complete treatment of kinetic-energy density in the exchange and correlation functionals.

We begin with the PBE (Ref. 11) exchange functional $F_{X\sigma}^{\text{PBE}}$. This is a theoretically sound starting point because it satisfies the correct uniform electron gas (UEG) limit and also has a reasonable form at large values of the reduced density x_σ (which equals $|\nabla\rho_\sigma|/\rho_\sigma^{4/3}$), which is important for treating noncovalent interactions.^{19,27,28} The exchange energy is then approximated as

$$E_X^{(0)} = \sum_{\sigma} \int d\mathbf{r} F_{X\sigma}^{\text{PBE}}(\rho_\sigma, \nabla\rho_\sigma) \left(\sum_{i=0}^m a_i w_{\sigma}^i \right), \quad (1)$$

where w_σ is the function of τ_σ suggested by Becke¹⁴ because it reduces to 0 for a uniform electron gas, but becomes negative in regions of delocalized one-electron exchange holes.

^{a)}Electronic mail: truhlar@umn.edu

The static correlation contribution to the exchange functional must be different in such negative- w_σ regions, typically associated with fractionally occupied single orbitals, than in more typical regions where $w_\sigma \approx 1$. Since we do not know the dependence on w_σ *a priori*, we take the coefficients a_i as shaping parameters. We require $a_i=1$ to constrain the exchange functional to have the correct UEG limit. The power series in Eq. (1) is similar to that employed by Becke¹³ in an earlier work with older expressions for $F_{X\sigma}$ and w_σ , but we changed the form and use more terms in the series. The motivation here is that combining a very flexible representation of inhomogeneity with a well-validated function of ρ_σ and $\nabla\rho_\sigma$ can lead to a more accurate exchange functional that corrects the treatment of delocalization, thereby requiring less cancellation of errors by the correlation functional. However, as first pointed out by Becke,⁷ the density-based exchange is still not correct for weak electron-electron coupling, and this can be corrected^{2,7,9,14,17} by mixing in a small percentage X of HF exchange:

$$E_X = \left(1 - \frac{X}{100}\right) E_X^{(0)} + \frac{X}{100} E_X^{\text{HF}}. \quad (2)$$

The correlation energy is the sum of opposite-spin and same-spin components:^{9,12,21,24–28}

$$E_C = E_C^{\text{opp}}(\rho_\sigma, \nabla\rho_\sigma) + \sum_{\sigma=\alpha,\beta} E_C^{\sigma\sigma}(\rho, \nabla\rho_\sigma, \tau_\sigma).$$

The form of the opposite-spin functional is the same as that used in the τ -HCTH (Ref. 24) and BMK (Ref. 26) functionals, and it contains six parameters $\gamma_{C\alpha\beta}$ and $c_{C\alpha\beta,i}$ with $i=0, 1, 2, 3$, and 4. The parallel-spin term is based on the work of Becke,^{9,12} who introduced a parameter⁹

$$D_\sigma = 2 \left(\tau_\sigma - \frac{|\nabla\rho_\sigma|^2}{8\rho_\sigma} \right)$$

and used $D_\sigma/D_\sigma^{\text{UEG}}$ to remove the self-correlation error. In the present work we replace $D_\sigma/D_\sigma^{\text{UEG}}$ by the variable $D_\sigma/2\tau_\sigma$ (also suggested by Becke¹³ who relabels $2\tau_\sigma$ as τ_σ) because it eliminates the integration grid instabilities noted by Johnson *et al.*³² The form of the $\nabla\rho$ dependence of $E_C^{\sigma\sigma}$ is the same as that in the τ -HCTH and BMK functionals and involves six parameters $\gamma_{C\sigma\sigma}$ and $c_{C\sigma\sigma,i}$ with $i=0, 1, 2, 3$, and 4. For both E_C^{opp} and $E_C^{\sigma\sigma}$, the $\nabla\rho$ dependence differs from the original formulation of Becke¹² by increasing the number of terms in a power series from three to four, and in that we require $c_{C\alpha\beta,0}=c_{C\sigma\sigma,0}=1$. In agreement with the philosophy of the B95 functional,⁹ this forces the correlation functionals to have the correct UEG limit, which is not enforced in a considerable body of work^{12,21,23,24,26} using similar correlation functionals. (The new correlation functional also differs from the τ -HCTH and BMK functionals by including the $D_\sigma/2\tau_\sigma$ factor for the parallel-spin case.) In addition to giving the correct UEG limit, our correlation functionals have two other theoretically important features: a rational polynomial in $\nabla\rho$ that accounts for the leading inhomogeneity correction to the correlation energy and a τ -dependent factor that eliminates self-correlation energy.

TABLE I. Optimized parameters (unitless).

a_i	1.0, 0.081 51, -0.439 56, -3.224 22, 2.018 19, 8.794 31, -0.002 95, 9.820 29, -4.823 51, -48.175 74, 3.648 02, 34.022 48
$\gamma_{C\alpha\beta}$	0.0031
$c_{C\alpha\beta,i}$	1.0, 3.773 44, -26.044 63, 30.699 13, -9.226 95
$\gamma_{C\sigma\sigma}$	0.06
$c_{C\sigma\sigma,i}$	1.0, 3.785 69, -14.152 61, -7.465 89, 17.944 91
X	28

Following Becke,¹³ we preoptimized the γ parameters to the correlation energies of He and Ne in a preliminary fit. All other parameters were optimized with a genetic algorithm to minimize the training function

$$F = \text{RMSEPB(AE6)} + \text{RMSE(IPEA8)} + \text{RMSE(K9)} \\ + 10\text{RMSE(NB4)} + 0.2\text{RMSE(AAE5)} \\ + \text{RMSE(MBE3)}.$$

The first term is the root-mean-square error (RMSE) per bond in the atomization energies of propyne (C_3H_4), cyclobutane (C_4H_8), glyoxal ($\text{C}_2\text{H}_2\text{O}_2$), silane (SiH_4), silicon dioxide (SiO_2), and disulfur (S_2). The second term is the RMSE for the ionization potentials of C, Cr, Cu, O, and OH and the electron affinities of C, O, and OH. The third term is the RMSE for the forward and reverse barrier heights and energies of the reactions of $\text{H}+\text{OH} \rightarrow \text{O}+\text{H}_2$, $\text{OH}+\text{CH}_4 \rightarrow \text{H}_2\text{O}+\text{CH}_3$, and $\text{H}+\text{H}_2\text{S} \rightarrow \text{H}_2+\text{HS}$. The fourth term is ten times the RMSE for four noncovalent complexation energies: $(\text{H}_2\text{O})_2$, $(\text{CH}_4)_2$, $(\text{C}_2\text{H}_4)_2$, and $(\text{C}_2\text{H}_4)(\text{F}_2)$. The fifth term is 0.2 times the RMSE in the absolute energies of H, C, O, S, and Si. The sixth term is the RMSE of the bond energies of Cr_2 and V_2 and the Cr–C bond of CrCH_3^+ . Thus the training set consists of 35 data. References for the training set data are in the supporting information.

Geometries and the treatment of spin-orbit terms are explained in previous papers.^{28–30} During parametrization, we used the DZQ basis set²⁹ for the calculations involving transition metals and the MG3S basis set³³ for other calculations. We used $m=11$. The optimized parameters are in Table I.

To assess the accuracy of the resulting density functional, we tested the results against a much larger set of 231 data. To gauge the performance for main group thermochemistry we use the mean unsigned error (MUE) on 135 data consisting of the MGAE109/04 database of 109 main-group atomization energies, the IP13/3 database of 13 main-group ionization potentials, and the AE13/3 database of 13 main-group electron affinities; this is called total mean unsigned error (TMUE). For kinetics we use the average MUE of 38 chemical barrier heights and 19 energies of reaction of 19 hydrogen atom transfer reactions in the HTBH38/04 database; this is called average mean unsigned error (AMUE). For noncovalent interactions, the question of whether to include counterpoise (CP) corrections for basis set superposition error is controversial. Because it is important to obtain accurate results both in cases where it is used to improve performance and in cases (e.g., condensed phases) where it is impractical, and because CP corrections often overestimate

TABLE II. Mean unsigned errors in kcal/mol. (The four best entries in each column are in boldface.)

Method	Reference	Thermochemistry TMUE	Kinetics AMUE	Nonbonded interactions MMMUE	Metal binding MMUE	Straight average ^a	Weighted average ^b
$\rho_\sigma, \nabla \rho_\sigma, \tau_\sigma$ in E and C , HF in E							
M05	Present	0.99	1.39	0.52	6.53	2.36	0.43
TPSSh	25	1.37	4.31	1.05	9.45	4.05	0.83
$\rho_\sigma, \nabla \rho_\sigma, \tau_\sigma$ in C , HF in E							
PW6B95	28	0.81	1.99	0.62	13.87	4.32	0.61
B1B95	9	0.96	1.79	1.25	14.78	4.69	0.76
PWB6K	28	1.72	1.42	0.38	23.77 ^c	6.82	0.81
$\rho_\sigma, \nabla \rho_\sigma, \tau_\sigma$ in E , HF in E							
BMK	26	0.94	1.08	1.10	27.27 ^c	7.60	0.92
$\rho_\sigma, \nabla \rho_\sigma$, HF in E							
B97-2	23	1.02	2.16	1.23	7.55	2.99	0.65
B97-1	21	1.07	2.94	0.71	10.50	3.81	0.66
mPW1PW91	19	1.32	2.34	0.95	16.62	5.31	0.81
B3LYP	5, 17, and 18	1.41	3.09	1.13	12.56	4.55	0.83
$\rho_\sigma, \nabla \rho_\sigma, \tau_\sigma$ in E and C							
TPSS	15	1.36	5.12	1.22	7.41	3.78	0.87
$\rho_\sigma, \nabla \rho_\sigma, \tau_\sigma$ in C							
BB95	9	2.40	4.89	1.67	11.58	5.14	1.14
$\rho_\sigma, \nabla \rho_\sigma$							
BLYP	4 and 5	1.93	4.90	1.63	6.55	3.75	0.99
mPWPW91	19	2.24	5.20	1.30	8.57	4.33	1.01
PBE	11	3.01	6.01	1.14	9.53	4.92	1.13
G96LYP	5 and 8	2.31	4.22	2.75	6.44	3.93	1.21
ρ_σ							
SPWL	6	14.70	12.05	2.18	26.87	13.95	3.33
Average		2.33	3.82	1.23	12.93	5.08	1.00

^a(TMUE+AMUE+MMMUE+MMUE)/4.^b((TMUE/2.33)+(AMUE/3.82)+(MMMUE/1.23)+(MMUE/12.93))/4.^cFor PWB6K and BMK, this value was approximated as the average of the MUEs for TMAE4/05 and MLBE4/05 with the DZQ basis. For the other 15 cases, the MMUE for this smaller test set (Refs. 29 and 30) and smaller basis (eight data, eight calculations) agrees with that for the larger test set (31 data, 62 calculations) with an average absolute deviation of only 8.9%.

the correction, we measure the performance on noncovalent interactions by

$$\text{MMMUE} = [\text{MMUE}(\text{no-CP}) + \text{MMUE}(\text{CP})]/2,$$

where each MMUE is averaged over individual MUEs for five kinds of noncovalent interaction represented by the HB6/04, CT7/04, DI6/04, WI7/05, and PPS5/05 databases of noncovalent complexation energies for, respectively, six hydrogen bonding dimers, seven charge-transfer complexes, six dipole interaction energies, seven weakly bound dimers governed by dispersionlike interactions, and five π - π stacking complexes. The performance for metals is the average of the mean unsigned errors for the TMAE9/05 database of nine transition-metal dimer atomization energies and the MLBE22/05 database of 22 metal-ligand bond energies. This was further averaged over the DZQ and TZQ (Ref. 29) basis sets; this average is called MMUE. References for databases, counterpoise corrections, and software are in the supporting information.³⁴

Table II shows the results for the four error quantities and compares them with 16 of the most popular and best performing density functionals in literature. For this purpose the results are divided into groups depending on the variables included in the functional. For each of the four categories of interaction we have put the results for the four best performing functionals in boldface. The new functional, denoted M05 (for Minnesota 2005), is in the top 4 in all four categories, and it is the only functional to appear in the top 4 in more than two categories. The simultaneous good performance for kinetics (barriers) and metallic binding is especially encouraging; no previous functional with Hartree-Fock exchange has ever performed well for metals, and no functional without Hartree-Fock exchange has ever performed well for hydrogen-transfer-barrier heights. The second last column of Table II is an average of the four mean unsigned errors. The M05 functional has an average error 1.3 times smaller than the second best performing method (B97-2). Because the errors for metal binding are much larger than the

others, we also computed a weighted average where each error is divided by the average error of all 17 functionals for that quantity; this is shown in the last column. With this fitness function, the M05 functional performs 1.4 times better than the second best performing functional (which is now PW6B95).

We found that we could obtain very similar results (the average error increases to only 2.53 kcal/mol and the scaled average to 0.44) by employing the same procedure, with $m = 10$ and $X = 30$, and the PBE exchange functional replaced by the mPW (Ref. 19) one, which is based on the earlier B88 (Ref. 5) and PW91 (Ref. 10) functionals. Thus the treatment of kinetic-energy density and the optimization of the correlation functional for complementarily to a given exchange functional are the key ingredients in the new functional, not the precise choice of $F_{X\sigma}$.

We checked that we obtain relatively good results for dipole moments as well as dipole-bound complexes.³⁴

We conclude that the new treatments of the exchange and correlation functionals are well balanced in a way that allow accurate Hartree-Fock exchange to be included even for systems with large static correlation. For a broad accuracy across the four properties tested (nonmetallic and metallic thermochemistry of molecules, chemical reaction barriers, and noncovalent interaction energies), the new density functional outperforms all previously available density functionals by a significant margin.

This work was supported in part by the National Science Foundation and the U.S. Department of Defense.

¹W. Kohn, A. D. Becke, and R. G. Parr, *J. Phys. Chem.* **100**, 12974 (1996).

²J. Perdew, in *Density Functional Theory*, edited by P. Geerlings, F. DeProft, and W. Langenaeker (VUB, Brussels, 1999), pp. 87–109.

³J. P. Perdew, *Phys. Rev. B* **33**, 8822 (1986).

⁴A. D. Becke, *Phys. Rev. A* **38**, 3098 (1988).

⁵C. Lee, W. Yang, and R. G. Parr, *Phys. Rev. B* **37**, 785 (1988).

⁶J. P. Perdew and Y. Wang, *Phys. Rev. B* **45**, 13244 (1992).

⁷A. D. Becke, *J. Chem. Phys.* **98**, 1372 (1993).

⁸P. W. M. Gill, *Mol. Phys.* **89**, 433 (1996).

⁹A. D. Becke, *J. Chem. Phys.* **104**, 1040 (1996).

¹⁰J. P. Perdew, K. Burke, and Y. Wang, *Phys. Rev. B* **54**, 16533 (1996).

¹¹J. P. Perdew, K. Burke, and M. Ernzerhof, *Phys. Rev. Lett.* **77**, 3865 (1996); **78**, 1396(E) (1997).

¹²A. D. Becke, *J. Chem. Phys.* **107**, 8554 (1997).

¹³A. D. Becke, *J. Chem. Phys.* **109**, 2092 (1998).

¹⁴A. D. Becke, *J. Chem. Phys.* **112**, 4020 (2000).

¹⁵J. Tao, J. P. Perdew, V. N. Staroverov, and G. E. Scuseria, *Phys. Rev. Lett.* **91**, 146401 (2003).

¹⁶A. D. Becke, *J. Chem. Phys.* **122**, 64101 (2005).

¹⁷A. D. Becke, *J. Chem. Phys.* **98**, 5648 (1993).

¹⁸P. J. Stephens, F. J. Devlin, C. F. Chabalowski, and M. J. Frisch, *J. Phys. Chem.* **98**, 11623 (1994).

¹⁹C. Adamo and V. Barone, *J. Chem. Phys.* **108**, 664 (1998).

²⁰H. L. Schmider and A. D. Becke, *J. Chem. Phys.* **108**, 9624 (1998).

²¹F. A. Hamprecht, A. J. Cohen, D. J. Tozer, and N. C. Handy, *J. Chem. Phys.* **109**, 6264 (1998).

²²B. J. Lynch, P. L. Fast, M. Harris, and D. G. Truhlar, *J. Phys. Chem. A* **104**, 4811 (2000).

²³P. J. Wilson, T. J. Bradley, and D. J. Tozer, *J. Chem. Phys.* **115**, 9233 (2001).

²⁴A. D. Boese and N. C. Handy, *J. Chem. Phys.* **116**, 9559 (2002).

²⁵V. N. Staroverov, G. E. Scuseria, J. Tao, and J. P. Perdew, *J. Chem. Phys.* **119**, 12129 (2003).

²⁶A. D. Boese and J. M. L. Martin, *J. Chem. Phys.* **121**, 3405 (2004).

²⁷Y. Zhao and D. G. Truhlar, *J. Phys. Chem. A* **108**, 6908 (2004).

²⁸Y. Zhao and D. G. Truhlar, *J. Phys. Chem. A* **109**, 5656 (2005).

²⁹N. E. Schultz, Y. Zhao, and D. G. Truhlar, *J. Phys. Chem. A* **109**, 4388 (2005).

³⁰N. E. Schultz, Y. Zhao, and D. G. Truhlar, *J. Phys. Chem. A* (in press).

³¹T. A. Ziegler, *Chem. Rev. (Washington, D.C.)* **91**, 651 (1991).

³²E. R. Johnson, R. A. Wolkow, and G. A. DiLabio, *Chem. Phys. Lett.* **394**, 334 (2004).

³³B. J. Lynch, Y. Zhao, and D. G. Truhlar, *J. Phys. Chem. A* **107**, 1384 (2003).

³⁴See EPAPS Document No. E-JCPSA6-123-301543 for references for databases, counterpoise corrections, and software, for dipole moment results, for extra details on the errors in noncovalent interactions, and for ethylene dimer potential. This document can be reached via a direct link in the online article's HTML reference section or via the EPAPS homepage (<http://www.aip.org/pubservs/epaps.html>).